Moving to Edmonton Made Easy

Project Proposal

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INTRODUCTION

Edmonton is known for its large and diverse city; people from within and outside the city are keen on learning what's a good neighborhood to live in. It's not exactly easy to come to such a conclusion. There are many factors to consider, such as neighborhood safety, community ethnic population, schools, pricing, etc. The current system of looking for homes in Edmonton is quite cumbersome. A user must look at a realtor website like livrealestate.ca, which does a great job finding available homes and pricing, but it fails to consider other factors a user may care about. Therefore, the person will have to do additional research. They might have to go through multiple resources, such as google maps, social media, and word of mouth, to make an informed decision. This process can be very time-consuming. A person can be in a situation during the research process where the neighbourhood they are interested in checks many of their requirements. However, one aspect that turns them away is the high crime rate. This person perhaps spent hours researching the neighbourhood, and now they must restart the process. There is a clear need to streamline finding homes in Edmonton. We aim to help users find a good neighborhood to live in without the hassle of going through multiple resources to decide. We are primarily targeting immigrants or migrants, and we hope to help them find communities where they will feel at home through the languages spoken. Most users will have their most important questions answered through a data visualization application, drastically reducing the time spent searching for homes in Edmonton.

Problem & Solution

The Canadian government plans to welcome approximately half a million new residents into Canada for the years 2022 to 2024. This plan poses many problems for future Canadians, including searching for the perfect place to move in. Looking for the right place requires a lot of research, and most of the time, it is hard to find the right resource and information one would need to decide whether the area they are looking at would best fit their lifestyle.

We decided to put ourselves in their shoes by asking general questions one would ask if they were immigrating or migrating to a new area. Simple questions such as:

- "What is the crime rate around this area?"
- "What are the average house prices in this area?"
- "Are there good schools I could put my kids in?"
- "Will we "feel at home" or welcomed into this new place?"

These questions are planned to be addressed in this project. For this project we are outlining the answers to these questions to aid future Edmontonian immigrants and migrants with their research.

With these questions in mind, we plan to design an interactive web application to help target users explore neighborhoods in Edmonton. The web app will contain a map of Edmonton, menus, and information visualizations that will display relevant information such as crime rate, property assessments, school locations, and spoken languages in the selected area.

As stated above, our target users will be future Edmontonian immigrants and migrants. Our proposed solution will help users decide whether Edmonton is a place they want to move to or not. The application will allow users to see information regarding a specific neighborhood or ward in Edmonton. Information such as commonly spoken languages in neighborhoods could assist them to see if that community includes households that speak their language. Knowing this kind of information could help them find others they could relate to the most, possibly aiding their transition to their new home. Another possible target user is the City of Edmonton. The city could use this application to assess property values, study the demography of the neighborhoods, and crime rates. Perhaps knowing where high concentrations of violent crimes occur could assist city officials to decide where they

should place more enforcements. Knowing the housing statistics of Edmonton, the city could also help future house owners to decide which area fits their budget the most.

The application's main feature will consist of a map of Edmonton which users could interact with by selecting neighborhoods or wards they wish to research. When a user selects an area, statistics such as crime rate, language spoken, school location, and property assessment will be displayed through visualizations. Menus and searching features will also be included. These features will allow users to research specific information such as property assessment for a certain neighborhood, or most spoken language in an area. The datasets that will be used are obtained from Canadian government open data sites such as data.edmonton.ca, so the data shown is as accurate as what the city of Edmonton provides.

Moving to a completely different community is daunting, especially if you know nothing about the region. Our goal for this project is to help target users to decide whether Edmonton could be the best fit for their new home.

Methodology

As this project's scope will be to create a multi-layered information visualization tool, we will combine several datasets to give the user a complete understanding of their zone of interest. The primary information visualization tool will use three data sets gathered from the City of Edmonton Open Data Portal and the Edmonton Police Services website. The first data set will contain information on crime occurrences and their exact locations provided, separated by categories of crimes as well as grouped in different tiers within each given category. Given that the Edmonton Police Service provides this data set, it has high reliability, and since it is updated daily, it provides a current snapshot for the entire city. Secondly, a data set on the assessed values for Edmonton properties produced by the City of Edmonton will be used to calculate the average price of homes for each neighbourhood and provide the user with a broad view of the area. The third data set used contains information about the languages (apart from English) spoken in every given Edmonton neighbourhood.

The dataset on crime occurrences provides dates and several types of crimes committed. Most importantly, it provides the location's latitude and longitude, which will be used to display the particular crime on a map. This visualization will be beneficial when the user selects a specific neighbourhood they want to focus on. A drawback in this dataset is a lack of detailed neighbourhood information, which would tie each crime occurrence to a neighbourhood rather than just a coordinate. As a solution, a separate method must be introduced, determining whether a given coordinate is within the borders of a particular neighbourhood. Regardless of this downside, the data is well categorized, and the general information is well distributed into specific columns.

The property assessment dataset provides a broad view of each neighbourhood, and its average assessed value, giving the user an idea about what prices they can expect to find in any target neighbourhood. We will achieve this feature by filtering only the residential properties to provide an accurate view. This dataset contains properties that do not have a neighbourhood listed; therefore, they must be filtered out for accuracy.

The language variety dataset provides information about the number of dwellings in each neighbourhood that speak another language besides English. It lists the ten most spoken languages with "Other" having grouped the remaining languages. Although the particular columns well represent

the data, the year this information was last updated was 2016, providing a possibly outdated demographic. Despite this, the assumption is that most people do not move very often, and the data is still relevant for a user interested in a particular area.

To complete the vision for this project, the principal language will be Python. Through the use of Python, we will have access to valuable libraries like Pandas and Plotly. Pandas offers excellent tools and methods to manipulate and assess data quickly and effectively for the back-end section. For the visualization part of the project, Plotly has developed an excellent framework in Dash that will allow us to display the intended information and make it intuitive for users to customize the information to their particular interests.

This project offers several expansion opportunities depending on the data sets available. Upon project completion, the scope and information it provides can be easily updated for any major city with similar data sets. Furthermore, additional data sets on school information can be added as another layer, helping to guide users in their decision-making. Lastly, we will be using static datasets, but through the use of APIs and database hosts, this project could be updated to refresh the data every time the user accesses it, making it more current.

Design

Our proposed solution will look to help new Edmontonians decide on housing in a neighborhood by looking through crime statistics, housing assessments, school areas, and languages spoken. Implementing this solution would involve features available on the Dash framework to create a simple-to-use web-based dashboard. We are looking to implement a search with filters for each dataset to assist users in finding information related to each locational area. These features allow users to filter crime statistics, housing assessments, spoken languages, and schooling areas based on wards and neighbourhoods. The filters could also include specified ranges and the minimum, average, and maximum values for each region, including a total area. Visualizing our information will be done through the graphing library Plotly. Locational data will be represented through maps, and the hard data could be represented in text and visualized as graphs. Our reason for this design decision is that users can easily search through the dashboard and receive immediate information.

We propose a dashboard system for displaying the frequency of crimes, the average housing values for each target neighbourhood, and the spoken language in each community. Our layout will have an interactive city map with filters for wards and neighbourhoods and a search bar for specific addresses. A sidebar with a drop-down menu for each crime data category, housing assessments, and spoken language will change the map based on the user selection. Further down the screen, other graphs will be displayed, such as the percentage of specific crimes and spoken languages in each ward and neighbourhood, helping the user make an informed decision.

Project Timeline

Milestone 1:

Oct. 13 - Oct. 14: Complete Project Proposal & Summarize Data

Milestone 2:

Oct. 14 - Oct. 18: Work on two low-fidelity prototypes in groups of two

Oct. 15 - Oct. 24: Exploratory Data Analysis individual work (including graphs)

Oct. 25: Deliverable and presentation: Storytelling with data.

Oct. 25 - Oct. 31: Finalize low fidelity prototypes and work on design document

Nov. 1: Present prototypes in class. Choose one prototype to model project on

Nov. 1 - Nov. 4: Finalize Design document and submit.

Nov. 4: Initialize web application features such as the dashboard and buttons, while continuing with producing required visualizations such as graphs

Nov. 4 - Nov. 11: Add functionalities to the dashboard features including maps, menus, and backend connections

Nov. 11 - Nov. 25: Keep working on the application functionalities and required visualizations

Nov. 25 - Dec. 6: Finalize project, touching up on the features, graphs, visualizations, and design

Final Deliverable:

Dec. 6: Interactive Dashboard Application presentation and Demo. Project submission.